

Osama Dabbousi

GPU algorithm design • High-performance computing • Scientific simulation

osama.dabbousi@gmail.com • osamadabb.github.io • Thuwal, Saudi Arabia

Education

University of Oxford

DPhil in Engineering Science — Rhodes Scholar

Oct. 2026 – (incoming)
Oxford, United Kingdom

- Advised by Prof. Paul J. Goulart and Prof. Wes Armour.

King Abdullah University of Science and Technology (KAUST)

M.S. in Computer Science, GPA 4.0/4.0 — Advisor: Prof. Matteo Parsani

2026
Thuwal, Saudi Arabia

Boston University

B.S. in Data Science, Summa Cum Laude — Faculty of Computing & Data Sciences

2024
Boston, MA

Selected coursework: High-Performance Computing, Numerical Linear Algebra, Concurrency, Digital Design & Computer Architecture, Computer Graphics, Databases, Artificial Intelligence.

Honours & Awards

- **Rhodes Scholar-Elect** (2025) — full scholarship for graduate study at the University of Oxford, awarded for academic merit, leadership, and service.
- **CDS Academic Excellence Award**, Boston University (2024) — recognised as the top undergraduate in the Faculty of Computing & Data Sciences.
- **KAUST Gifted Student Program Scholarship (KGSP)** (2020–2026) — fully funded, merit-based scholarship for top Saudi students, covering undergraduate study abroad and the M.S. at KAUST.

Research Interests

GPU algorithm design and architecture, sparse linear algebra, high-performance and parallel computing, scientific simulation, and algorithm–hardware co-design.

Research Experience

Advanced Algorithms & Numerical Simulation Lab, KAUST

Graduate Researcher — Advisor: Prof. Matteo Parsani

Oct. 2024 – Present
Thuwal, Saudi Arabia

- Designed and built **ASTRA-GBT**, a GPU-accelerated Gaussian Beam Tracing framework for acoustic propagation in inhomogeneous, moving media (M.S. thesis).
- Implemented the full pipeline in Numba CUDA with CuPy and MPI multi-node parallelism, including a checkpoint-and-replay architecture and a custom CUDA ray–surface intersection routine.
- Scaled to 2,048 GH200 GPUs on the Shaheen III supercomputer, achieving ideal weak scaling and 91% strong-scaling efficiency at 128 GPUs, on a city-scale acoustic simulation of the King Abdullah Financial District.
- Contributed to porting an in-house CFD solver from CPU to GPU, focusing on memory-hierarchy optimisation and coalesced memory access.

Machine Learning Research Group, Boston University

Undergraduate Researcher — Advisor: Prof. Mark Kon

Sept. 2023 – Aug. 2024
Boston, MA

- Co-developed, with a team of eight, a meta-learning pipeline that automatically selects among dozens of training configurations to optimise cancer-classification accuracy.
- Built the distributed, multi-node training backend (SSH-orchestrated) and the core modules for data augmentation, feature extraction, and model training.

Teaching

Teaching Assistant — Boston University & KAUST

Jan. 2023 – Present

- Taught four quantitative courses across two institutions: Introduction to Algorithms, Natural Language Processing, Probability & Statistics, and Numerical Linear Algebra.
- Led discussion sections of 10–20 students and held office hours, giving detailed feedback on problem sets and projects.

Industry Experience

Aramco Americas

Boston, MA

Data Science Intern

June – Aug. 2023

- Built a retrieval-augmented generation (RAG) pipeline that queries thousands of academic articles to return well-cited, accurate answers.
- Designed an LLM-backed interface returning cited excerpts, and shipped it as a web application for company employees.

Data Science Intern

June – Dec. 2022

- Combined computer vision and NLP to build a semi-supervised pipeline that scraped and labelled geological thin-section images from academic articles, producing a dataset on the order of tens of thousands of samples.
- The resulting method was patented (U.S. Patent US-12579791-B2).

Selected Projects

Large-scale Dataset Compression with SVD

Python, PyTorch

SVD- and quantisation-based compression of the large-scale Red Sea temperature dataset, reducing it from 15.4 GB to 865 MB.

ChessGAN (*B.S. Capstone*)

Python, PyTorch

An adversarial architecture that trains a chess agent to play more human-like by penalising non-humanlike moves.

Raytracing from Scratch

C++

A ray tracer written in C++ with no external libraries, handling shading, shadows, reflection, and object motion and rotation.

Patents

- U.S. Patent **US-12579791-B2**, “Automated Methods for Generating Labeled Benchmark Data Set of Geological Thin-Section Images for Machine Learning and Geospatial Analysis.” Granted 2026.

Technical Skills

Programming: Python, C++, CUDA, SQL, Bash

HPC & GPU: Numba, CuPy, MPI, OpenMP, SLURM

Tools & libraries: PyTorch, Git, Linux

Languages: English (fluent), Arabic (fluent)